PROGRAM {FIRST FIT}

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#include <stdio.h>
#include <stdlib.h>

#define MAX 50

struct BLOCK {
    int size;
    int isallocated;
    int pallocated;
    int psize;
    int frag;
} block[MAX];

int main() {
    int nb, np, p_size[MAX];

    printf("Enter the no: of blocks\n");
    scanf("%d", &nb);

    printf("Enter the size of the blocks\n");
    for (int i = 0; i < nb; i++) {
        printf("Block%d: ", i);
        scanf("%d", &block[i].size);
        if (block[i].size < 0) {
            printf("Invalid Input\n");
            exit(0);
        }
        block[i].isallocated = 0;
        block[i].pallocated = -1; // -1 indicates no process
        block[i].psize = 0;
        block[i].frag = block[i].size;
    }

    printf("Enter the no: of processes\n");
    scanf("%d", &np);

    printf("Enter the size of the processes\n");
    for (int i = 0; i < np; i++) {
        printf("Process%d: ", i);
        scanf("%d", &p_size[i]);
        if (p_size[i] < 0) {
            printf("Invalid Input\n");
            exit(0);
        }
    }

    // Allocation logic: First Fit
    for (int i = 0; i < np; i++) {
        for (int j = 0; j < nb; j++) {
            if (block[j].isallocated == 0 && block[j].size >= p_size[i]) {
                block[j].isallocated = 1;
                block[j].pallocated = i;
                block[j].psize = p_size[i];
                block[j].frag = block[j].size - p_size[i];
                printf("Process%d is allocated to block%d\n", i, j);
```

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            break; // Move to the next process once allocated
        }
    }
}

// Display Output
printf("\nBlock\tIsallocated\tProcess_allocated\tBlockSize\tPSize\tFragment\n");
for (int i = 0; i < nb; i++) {
    printf("%d\t%d\t\t", i, block[i].isallocated);
    if (block[i].isallocated == 1)
        printf("%d\t\t\t", block[i].pallocated);
    else
        printf("-\t\t\t");

    printf("%d\t\t%d\t%d\n", block[i].size, block[i].psize, block[i].frag);
}

return 0;
}
```

OUTPUT {FIRST FIT}

```
blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_8$ gcc firstfit.c
blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_8$ ./a.out
Enter the no: of blocks
4
Enter the size of the blocks
Block0: 20
Block1: 30
Block2: 40
Block3: 50
Enter the no: of processes
5
Enter the size of the processes
Process0: 45
Process1: 25
Process2: 15
Process3: 35
Process4: 28
Process0 is allocated to block3
Process1 is allocated to block1
Process2 is allocated to block0
Process3 is allocated to block2

Block   Isallocated     Process_allocated       BlockSize       PSize   Fragment
0       1               2                       20              15      5
1       1               1                       30              25      5
2       1               3                       40              35      5
3       1               0                       50              45      5
blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_8$
```